TRACK 02 THE AGENT THAT CAN EXPLAIN WHY · OPEN AGENT HACKATHON 2026

FINMOZG
Variance Detective

Agents that tell a controller **why** gross margin moved, and prove every driver with rows from four systems, through Zetaris.

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github.com/ctmakc/finmozg-variance-detective

35.9% → 29.2%

−6.78 pp gross margin, September

"Why?" lives in four systems that share no keys.

GENERAL LEDGER	says **what** moved
PURCHASING / AP	says **which vendor, at what price**
ORDER EXPORT	says **which customers and products**
BANK FEED	says **what cash actually left**

Users: the controller or fractional CFO of a 20–200-person company at month-end close, and the owner who reads the memo. Today: 1–2 days of spreadsheet exports per close, and booking errors mistaken for real costs.

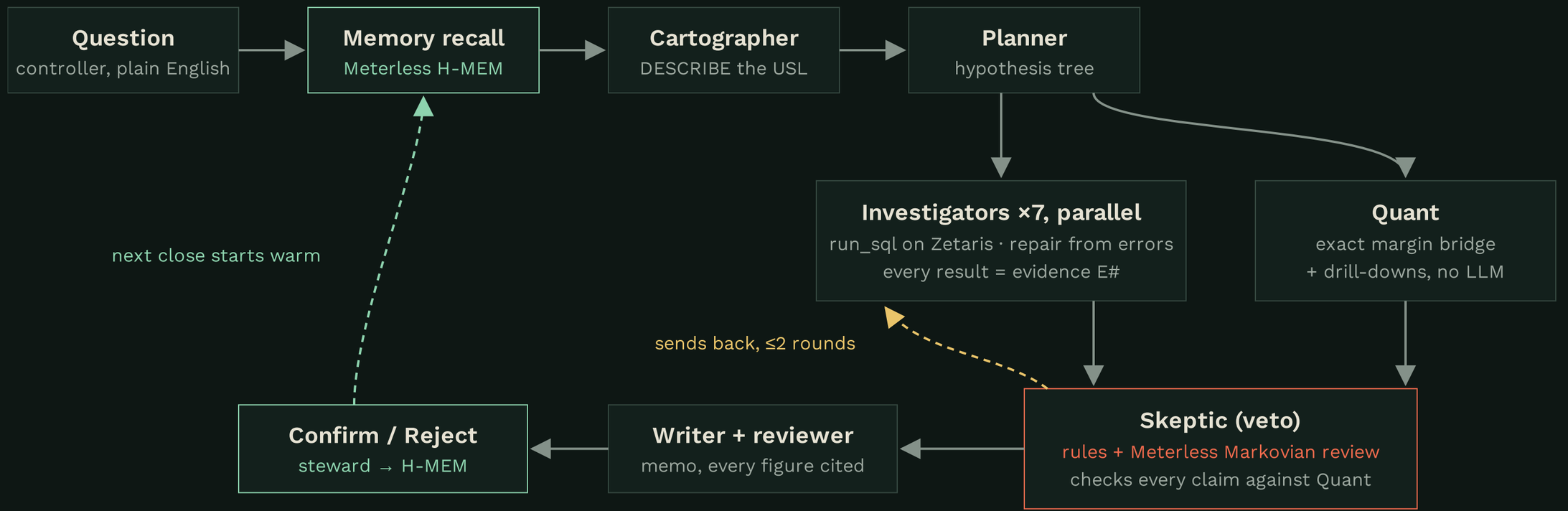
Ask in plain English. Get a memo where every number opens to its rows.

A team of agents plans hypotheses, queries all sources through one Zetaris semantic layer, computes an exact margin bridge, and lets a Skeptic veto anything without evidence.

DRIVER	CAD	PP	NAMED BY THE AGENTS
Booking error	−12,450	−3.46	MRP-88213 booked twice, paid once
Supplier price	−9,116	−2.54	Sierra Nut Co. above contract
Customer mix	−6,517	−1.81	Capital Wholesale 1,102 → 2,353 units
USD/CAD	−425	−0.12	measured, immaterial
Volume	+18,873	+1.26	more units sold
Residual	0	0.00	shown openly

~~Marketing spend rose~~ opex sits below gross margin

Actions in the memo: reverse the duplicate entry, claim the over-contract amount from the vendor



Meterless World Model: one shared evidence graph for every agent

Question ← Hypothesis ← Finding ← Evidence → Vendor · Invoice · Customer · each write attributed to the agent that made it

4 TECHNICAL STACK

DATA	**Zetaris** local platform · 2 Postgres (JDBC), CSV file store, 2 REST sources (bank feed, live Bank of Canada) · Unified Semantic Layer `finance_usl`
POWER	**Meterless Runtime** (Nightshift) as the primary AI provider, contract v4 · **NVIDIA Nemotron 3 Super** on a Runtime lane
COGNITION	**Meterless engines** H-MEM, World Model, Markovian (vendored, conformance 13/13, 8/8, 8/8)
APP	TypeScript · Node 20 · zod-validated model JSON · vitest (37 tests) · node:http API with server-sent events · vanilla web UI
BUILD	**Cursor** project and Cursor Agent CLI sessions, logged in `docs/cursor-sessions.md`
JUDGE PATH	`SAMPLE_MODE=true` replays recorded live runs · Docker image (CPU) · `POST /run` · `npm run eval` against planted ground truth

Zetaris is the only door to the data

- ✓ 5 sources registered by script; agents query only POST `/sql-editor/sqls/run`
- ✓ `finance_usl`: 9 business tables, PK/FK, activated from raw sources, BoC rate joined on invoice date
- ✓ `RUN DQ`: PK rule flags MRP-88213 (31 records, 2 invalid) before any agent asks
- ✓ Investigators write Spark SQL and repair it from Zetaris error messages

Meterless is memory, shared state and power

- ✓ **H-MEM**: schema map (warm run skips discovery), confirmed findings, Skeptic rulings, audit ledger
- ✓ **World Model**: one evidence graph for 10+ agents, provenance per write
- ✓ **Markovian**: Skeptic reviews evidence rows in bounded chunks with carryover
- ✓ **Runtime**: every model call app → Meterless → Nemotron, signed receipt per call

6 WHAT WAS UNIQUE

SKEPTIC FIRST	Deterministic veto against the Quant bridge and drill-down; it also **challenges false negatives**: "you found nothing, but the bridge measures CAD 6,517 for mix: find the rows".
DQ AS EVIDENCE	Zetaris USL data-quality rules are part of the investigation: the duplicate is a primary-key violation in the semantic layer.
CROSS-SYSTEM PROOF	A cost claim needs AP *and* bank; the memo shows "booked twice, paid once" with both row sets.
MEMORY THAT CHANGES THE NEXT CLOSE	A confirmed finding becomes the first check next month (H-MEM), visible in the UI.
MEASURED, NOT CLAIMED	Planted ground truth + 22-check eval: 16/22 on the first run → 22/22 after the Skeptic and drill-down changes.

Why did gross margin fall in September vs August?

35.93% → 29.15%

−6.78 pp · GM −10,034 CAD

gross margin 2026-08 → 2026-09 revenue 319,555 → 359,444 run r-20261022-013640-c8ea

600 s · SQL 61 (0 repaired) · model calls 64 served by nemotron-3-super-ollama via Meterless Runtime

C **MARGIN BRIDGE** 2026-08 → 2026-09 · CAD · pp of margin

DRIVER	CAD	PP	TICKS	EFFECT ON GROSS MARGIN
Gross margin 2026-08 35.93% of revenue	114,821			
Volume	+18,873	+1.26	✓ ⊕ Σ	
Customer / product mix wholesale	−6,517	−1.81	✓ ⊕ Σ	
Selling price	0	0.00		
Supplier unit price VEND-SNC · NUT-CAS	−9,116	−2.54	✓ Σ	
USD/CAD exchange rate VEND-SNC · VEND-LOI	−425	−0.12	✓ Σ	
Other goods cost	0	0.00		
Booking error (duplicate invoice) VEND-MRP · PKG	−12,450	−3.46	✓ ⊕ Σ	
Packaging spend	−399	−0.11	✓ Σ	

VOLUME	2026 to 43,721 in September 2026, raising volume but reducing gross margin by approximately CAD 10,034 due to lower per-unit contribution. 3 evidence · Investigator-h2 · round 2	+18,873	✓ ⊕ Σ	Confirm Reject
BOOKING ERROR	Vendor VEND-MRP invoice MRP-88213 for packaging (SKU PKG) was booked twice in AP/GL in 2026-09 (two entries of 12,450 CAD each) but paid only once (12,450 CAD), resulting in an extra packaging cost of 12,450 CAD that reduces gross margin. 3 evidence · Investigator-h8 · round 1	−12,450	✓ ⊕ Σ	Confirm Reject

Skeptic: Evidence rows support it and the amount reconciles with the bridge (CAD -12450).
Markovian review: supported in 1 chunk(s) · AP shows two bookings of 12,450 CAD each while bank shows only one payment of 12,450 CAD, indicating an unmatched duplicate booking that adds the claimed extra cost.

E19 · Investigator-h8 · 1 rows · finance_usl.vendor_bill · systems: AP · query 90da6eb0

```
SELECT vendor_invoice_no, COUNT(*) AS bill_count, COLLECT_SET(ap_doc_id) AS ap_doc_ids
FROM lightning.metastore.finmozg.finance_usl.vendor_bill
GROUP BY vendor_invoice_no
HAVING COUNT(*) > 1
```

vendor_invoice_no	bill_count	ap_doc_ids
MRP-88213	2	["AP-00031","AP-00030"]

E21 · Investigator-h8 · 1 rows · finance_usl.vendor_payment · systems: Bank · query 530889bc

```
SELECT vendor_invoice_no, COUNT(*) AS payment_count, COLLECT_SET(txn_id) AS txn_ids
FROM lightning.metastore.finmozg.finance_usl.vendor_payment
WHERE vendor_invoice_no = 'MRP-88213'
GROUP BY vendor_invoice_no
```

vendor_invoice_no	payment_count	txn_ids
MRP-88213	1	["BNK-00030"]

E24 · Investigator-h8 · 2 rows · finance_usl.purchase_line · systems: AP · query 461c157c

```
SELECT purchase_line_id, vendor_invoice_no, sku, qty, unit_price_orig, currency, fx_rate,
unit_cost_cad, amount_cad
FROM lightning.metastore.finmozg.finance_usl.purchase_line
WHERE vendor_invoice_no = 'MRP-88213'
```

8 FUTURE ENHANCEMENTS

- → Connect real ledgers through Zetaris (QuickBooks, Xero, NetSuite exports, bank APIs) and grow `finance_usl` per client
- → Inventory-aware costing (FIFO / weighted average) and budget-vs-actual bridges
- → Draft the fix: reversing journal entry and vendor credit request, queued for the controller's approval
- → Close calendar: run the detective nightly in the last week of the month, alert on new material drivers
- → Multi-entity and FX translation; more question types with their own Skeptic rules
- → Ship inside FINMOZG, our finance product for small companies